Justin Yu

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Education

University of Pennsylvania — Philadelphia, PA

BS/MS in Electrical Engineering *(Accelerated Program)* — *May 2028*

Skills

Hardware: Cadence Virtuoso, Spectre, Pegasus DRC/LVS, Quantus, SystemVerilog, Altium Designer, LTSpice, Oscilloscope, Wafer Probe
Software: C, C++, Python, Lua, Java, Linux/Unix Shell, Linux Kernel Internals, Git

Experience

AWS Annapurna ML Labs — Austin, TX

Hardware Development Engineering Intern — *May 2026 – August 2026*

- Validated power-delivery margins enabling a **10%** cut in **HBM**-rail power for a new **Trainium 3** packaging approach (zero failures over **36 hours**); built a telemetry pipeline cutting shmoo runtime from **4h to 45min**
- Root-caused an **HBM4 PHY** training failure, validating **17%** above rated speed with clean **DRAM BIST**; saved **2–4 weeks** of vendor debug
- Built an automated **RTL** testbench for an **HBM4** controller generating **60 JEDEC IDD7R**-style power-stress patterns in **4 seconds**, adopted for ongoing validation
- Characterized substrate **PDN** for **UCIe** and **HBM** rails via **wafer probe** before die-attach, matching the PDN model

Penn Electric Racing — Philadelphia, PA

Electrical Hardware Lead — *September 2024 – Present*

- Lead **30+** engineers across **20+** PCBs through a **RWD**-to-**4WD** redesign: higher **CAN** bandwidth, decentralized data acquisition, larger **LV**/**HV** packs
- Co-designed and routed the 4-layer **BMS** PCB for a **600V** tractive system with **10mm** HV/LV separation, exceeding FSAE rules; validated **6** critical boards

Computer Architecture Teaching Assistant — Philadelphia, PA

University of Pennsylvania — *August 2026 – Present*

- TA for **CIS 5710** (graduate): out-of-order execution, caches and memory hierarchies, virtual memory

UPenn Computational Intelligence Lab — Philadelphia, PA

Research Intern — *June 2025 – August 2025*

- Co-authored **ISCA 2026** paper on R2D2, a robot-reconfigurable datacenter network with direct optical links between disaggregated nodes: **sub-15 s** reconfiguration, **70%** less power than switching

Projects

Full-Custom Multiply-Accumulate (MAC) Unit | *Cadence Virtuoso, FinFET, Spectre*

- Designed a transistor-level MAC with two on-chip **6T SRAM** macros, a **4×4** array multiplier, **12-bit** accumulator, custom flip-flops, and an FSM sequencer; verified end-to-end dot products
- Laid out **DRC/LVS-clean** SRAM bitcell and 16×4 array with **Quantus** parasitic extraction; sized critical path with **logical effort** and characterized setup/hold, t_{cq}, $T_{CLK,min}$, and energy per dot product

Pipelined RISC-V Processor | *SystemVerilog, FPGA*

- 6-stage **RV32IM** pipeline with **AXI-Lite** memory, forwarding, and hazard stalling; synthesized to FPGA

PennOS | *C, Operating Systems*

- UNIX-like OS with a **priority round-robin scheduler**, **FAT** file system, and **POSIX** signals across 64 processes